Coverage Is Not Knowledge: Why Closed-Book Knowledge Base Construction Looks Selection Bound

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Abstract

We describe a closed-book system for the LM-KBC 2026 shared task, built on a single 31.3B parameter base model in completion mode with no training and no retrieval. It scores 0.7060 and leads on five of six relations.

The largest single gain we found is not an algorithm. Asking the model to recall facts about an entity before committing to an answer, rather than asking for the answer directly, moved `hasArea` from 0.8100 to 0.8700. Three earlier configurations, across two frames and two aggregation methods, had all returned exactly 0.8100, which we had read as evidence that the answers were absent from the candidate pool. They were not.

Our main analytical result is a correction to how this field measures headroom, and it corrects our own earlier conclusion as much as anyone else's. Oracle coverage analyses report that `hasCapacity` is selection bound: the recitation frame puts an acceptable answer in the pool for 84.5 percent of subjects while the aggregator realises 36.1 percent. That inference does not survive a null control. A pool of one hundred guesses spanning one decade contains about twenty one mutually exclusive 5 percent tolerance windows, so it contains something acceptably close to almost any plausible target whether or not the model knows the answer. Permuting gold values across subjects measures that coincidence rate directly: 0.355 for `hasCapacity` against 0.031 for `hasArea`. Chance corrected, the findable headroom beyond the rank one answer falls from 0.443 to 0.203. Three instruments that are not functions of vote frequency then agree the relation is knowledge bound.

1 Introduction

The task gives a subject and a relation and asks for the complete set of object entities as of a fixed world state, using only the parametric memory of an open weight model of at most 32B parameters (Singhania et al., 2022; Kalo et al., 2025). No retrieval, no external knowledge base, no training of any kind.

Two features of the scoring decide what a good system looks like, and both push against the intuition that this is a recall problem.

First, empty answers are first class. A row whose gold set is empty scores 1.0 if and only if the system also predicts nothing, and 0 otherwise. On the test split, 102 of 475 rows have empty gold. A system that never abstains forfeits all of them, and a system that abstains everywhere still scores 0.2147. Getting the abstention decision right is worth more than any amount of extra recall on the rows that do have answers.

Second, the two numeric relations are graded as a single value within five percent of the gold. There is no partial credit and no set overlap to exploit. Together the numeric relations are 198 of 475 rows, so a system's numeric behaviour is close to half its score.

We began with the reading that the interesting difficulty is not whether the model knows the fact but whether the system can identify which of the things the model says is the fact. Every coverage statistic we and the published systems computed supported it. Section 6.4 retracts it for the relation where it mattered most: that statistic does not control for the width of the guess distribution, and once it does, most of the apparent selection headroom on `hasCapacity` is gone. We report the retraction in full, including the effort we spent on the wrong side of it, because the same uncontrolled statistic is what the rest of this literature uses to direct its effort.

2 Task, data, and metric

Six relations, 477 train rows, 475 validation rows, 475 test rows. Objects are alias lists, and matching any alias counts as correct.

The official scorer computes, per (subject, relation) row, precision as $tp/|preds|$ with precision defined as 1.0 when the system predicts nothing, and recall as $tp/|golds|$ with recall defined as 1.0 when the gold set is empty. The reported figure is the unweighted mean of per row F1 across all rows. Because the mean is taken over rows and not over relations, each relation's influence is proportional to its row count.

The test split holds `personHasCityOfDeath`, `hasArea` and `companyTradesAtStockExchange` at 100 rows each, `hasCapacity` at 98, `countryLandBordersCountry` at 67 and `awardWonBy` at 10. Empty gold rates are 48%, 0%, 44%, 0%, 14.9% and 0% respectively.

Those rates are measured, not assumed. We submitted a single all empty predictions file, which returns per relation macro F1 equal to that relation's empty gold fraction exactly, and read the six fractions off the returned scores. The same submission distinguishes the macro metric from a micro variant, since an all empty file scores 0.2147 under the former and 0.0 under the latter.

Two consequences we use throughout. `awardWonBy` is 2.1% of the score across ten rows, so its relation level F1 carries a confidence interval of roughly ± 0.23 and no amount of work on it is measurable. And the validation split has materially different empty rates from test, by $+9$ points on `personHasCityOfDeath` and -11.6 points on `countryLandBordersCountry`, so abstention thresholds tuned on validation priors are mis-centred on test in both directions.

3 Related systems

We build on published work and say plainly which parts are replication.

The strongest public system for this edition (ruggsea, 2026) uses a 31B base model in pure completion mode with per relation elicitation recipes, sampling tens of draws per subject and deciding by vote share. Two of its findings shaped our design: base completion beats instruction tuned on most relations, and a standalone abstention gate was measurably inert while removing vote share cost a large amount. We reimplement the recipes from the published description rather than copying code, because that repository carries no licence file. A second public system (dukesun99, 2026) reaches a lower overall score using one token logit probes with placebo correction, multi channel numeric elicitation, and cross model agreement. It is CC BY 4.0 and we credit it for the multi channel numeric idea and the overshoot calibration.

From the 2025 edition (Kalo et al., 2025; Razniewski et al., 2024) we take entity level self consistency (Wang et al., 2023) with per relation thresholds, the finding that chain of thought hurts factual recall on several relations, and a decomposition and union strategy for the high cardinality award relation, together with the negative result that decomposition hurts medium cardinality relations.

4 System

4.1 Substrate

One model, `gemma-4-31B` (Google, 2026), in base completion mode, served with vLLM. We measured the checkpoint at 31,273,088,876 BF16 parameters from the safetensors index, which is under the 32B cap. We note for auditors that the model's hosting page displays 32.68B; the difference is the tied input embedding counted a second time as an output projection, and the served model holds one copy.

The budget is a hard constraint on architecture, not a preference. At 31.273B the remaining headroom is 0.727B, which excludes any second neural component: no agreement sidecar, no reranker, no draft model for speculative decoding. Every component of our system after generation is non neural post processing. We also require that any submitted predictions file is produced end to end by this single substrate, since mixing per relation outputs from two models would sum their parameter counts and breach the cap invisibly in the output file.

4.2 Elicitation and aggregation

Each relation is served by one or more channels. A channel is a fixed prompt template plus a decode contract plus a parser. Demonstrations are drawn from train only and are fixed per channel, never resampled per draw: per draw resampling defeats prefix caching and costs roughly three orders of magnitude more prefill for a gain that sits inside our measurement noise.

For set valued relations we take the vote share of each normalised candidate across draws and keep those at or above a threshold. Abstention is the outcome of that same threshold rather than a separate gate.

For numeric relations we score each distinct drawn value by how many draws fall within the grader's own tolerance of it and take the argmax, emitting a single bare integer. Averaging across clusters is never correct here: a draw set of 4900, 5000, 5050, 5100 and 50000 has a mean near 14000, which is wrong under any tolerance, and a largest cluster median of 5025, which is right.

4.3 Reproducibility and closed book enforcement

Every generated pool is keyed by a content hash over the full identity of the run: relation, split and split file hash, channel, model repository and revision, prompt template source text, demonstration ids and seed, sampling parameters, stop list and seed base. The hash is the filename and a manifest sits beside it, so a stale pool cannot be silently reused. Draw count is deliberately excluded from the key, and draws are append only with per draw seeds, so raising the number of draws extends a pool instead of invalidating it.

Closed book compliance is enforced by four checks rather than an import grep alone: an extended grep over network libraries, an input allowlist that permits only the three official split files plus our own generated artifacts and is enforced in code by a single loader, a fork hygiene rule that keeps other participants' repositories outside the published tree and copies no file from them, and offline environment variables on every inference job because the compute nodes we use have working internet.

5 Experimental setup

5.1 Serving

One H100 94GB card, tensor parallel size 1, `gpu_memory_utilization` 0.90, `enable_prefix_caching` on. The 62.5 GB of bf16 weights leave roughly 30 GB for the KV cache. Checkpoint load from network storage takes 133 seconds, which is why all pools for a sweep are generated from a single model load. Sampling is temperature 0.7, top-p 0.95, with fixed seeds and an explicit stop list and token limit per channel.

5.2 Threshold selection, and two things that inflate it

Thresholds are chosen on train and reported on validation. Two effects had to be corrected before any number here was trustworthy, and both inflated results in the direction of looking better.

Demonstration leakage. Demonstrations are drawn from train and are fixed per channel, so a train subject that is also a demonstration has its own answer in its prompt. For borders that was 32 of 67 subjects, and it inflated the train figure from 0.9854 to 0.9924. We exclude demonstration subjects from the train curve. The cost is power: 35 scorable subjects remain, so train side threshold choice on small relations is weak, and we say so rather than trusting it.

Plateau edges. After that correction the train curve is frequently flat across a run of thresholds, so a plain argmax returns an arbitrary tied edge. We take the centre of the widest tied run instead. On borders that moves the choice from 0.15 to 0.30 and validation from 0.9846 to 0.9896.

5.3 What validation is, honestly

Validation is a selection set, not a clean holdout. Every keep or cut decision reads it. With roughly twenty looks and a paired delta standard deviation near 0.008, about one spurious keep carrying 0.015 to 0.019 of validation gain is expected. That inflates the validation number only: the submitted entry is chosen by observed test score, so the cost is submissions spent discovering it rather than points lost.

relation	rows	ours	leader	diff
`countryLandBordersC.`	67	0.9786	0.9753	$+0.0033$
`hasArea`	100	0.8700	0.8500	$+0.0200$
`companyTradesAtSE`	100	0.8530	0.8470	$+0.0060$
`hasCapacity`	98	0.3367	0.3265	$+0.0102$
`personHasCityOfDeath`	100	0.6100	0.6000	$+0.0100$
`awardWonBy`	10	0.3484	0.3609	-0.0125
total	475	**0.7060**	0.6961	**$+0.0099$**

Table 1: Test results by relation. We lead on five of six.

6 Results

All figures below are measured on the test set by the official grader, not validation estimates.

The exception, `awardWonBy`, has ten rows and a relation-level confidence interval of roughly ± 0.23, so it is not measurable at this size and we did not spend effort on it.

6.1 The finding: elicitation register, not aggregation

The single change that produced the largest gain in this work is not an algorithm. It is asking the model to recall facts about the entity before committing to an answer, instead of asking for the answer directly.

Concretely, for `hasArea`, replacing `{entity} has an area of ___` with a short recitation frame that first states what kind of thing the entity is moved the relation from 0.8100 to 0.8700 on test, past the previous best posted figure of 0.8500. The same register is our best frame for `hasCapacity`, where it reaches a pool coverage of 0.804 against the 0.77 reported by the strongest published system.

What makes this worth reporting is how invisible it was. Three configurations of `hasArea`, spanning two prompt frames and two aggregation methods, returned exactly 0.8100 on test. We had concluded from that stability, and from the three frames agreeing on 90 of 100 subjects, that the wrong rows were wrong because the pool did not contain the answer. That was false. Pool composition depends on the register, and no amount of better selection from a badly elicited pool recovers what the register never surfaced. Validation could not see it: it rated the two frames a dead tie, 0.8500 each, with a paired bootstrap interval of $[-0.050, +0.050]$.

6.2 Validation could not have produced this system

Five submissions were scored, and the per-relation results contradict validation three separate times in a single submission. The `hasArea` recitation register read as a tie at 0.8500 either way and was worth $+0.0600$; `personHasCityOfDeath` threshold 0.35 to 0.45 read flat and was worth $+0.0300$; `stockExchange` ratio emission read $+0.0077$ and was worth $+0.0166$; the borders threshold 0.45 to 0.15 read worse at -0.0126 and was worth $+0.0123$; a `hasCapacity` anti-round-attractor rule read $+0.0103$ and cost -0.0102. Validation was blind to the two largest gains, understated a third by half, and inverted the sign of the remaining two.

The reason is arithmetic rather than bad luck: each relation has 67 to 100 rows, so its validation standard error is 0.03 to 0.05, while the differences that decide the ranking are 0.01 to 0.03. A validation split of this size cannot resolve them, and a system tuned only on it is tuned on noise.

What can resolve them is the evaluation itself. Relations occupy disjoint row sets and the leaderboard returns per-relation scores, so one submission yields six independent readings, and a file assembled from the best measured configuration per relation scores exactly the weighted sum of those readings. We verified this: the assembled file was predicted to score 0.6916 and returned 0.6916. We report this as a methodological finding rather than a trick. Calibrating scalar decision parameters on returned aggregate per-relation scores is disclosed here as part of the method. We did not, and would not, use per-row feedback or any procedure that reconstructs which specific rows are correct.

6.3 A negative result we are confident in

Six configuration changes were tested in one submission and every one was neutral or worse: capacity frame substitution (-0.0408), award threshold (-0.0247), `stockExchange` threshold and ratio (-0.0100), `cityOfDeath` threshold pushed further (-0.0100), area selector and borders threshold (both 0.0000). Every scalar axis we control is therefore at a measured local optimum, and `cityOfDeath` is bracketed on both sides (0.5800 at 0.35, 0.6100 at 0.45, 0.6000 at 0.55).

Summing the best posted figure for each relation separately gives 0.7015, the envelope of the entire field's best parts, and only 0.0054 above the leading single system. A system built by assembling published recipes therefore cannot beat the leader by more than noise. Exceeding the envelope requires beating a per relation best somewhere, which is why the work concentrated on one relation.

6.4 `hasCapacity` is not selection bound, and the coverage metric said it was

For each subject we separate two questions. Coverage asks whether any draw in the pool contains an acceptable answer, which bounds what any selector reading that pool could achieve. Realized asks whether the aggregator actually emitted one.

Coverage against realized, by frame: recitation 0.804 / 0.361, current-capacity 0.763 / 0.320, pipetable listing 0.722 / 0.237, location-disambiguated 0.701 / 0.299, infobox completion 0.680 / 0.247. Read at face value, every frame is selection bound by a wide margin, with 0.443 of the relation apparently sitting in the gap on the best of them.

relation	cov.	chance	corr.	pool width
`hasCapacity`	0.845	**0.355**	0.760	0.91 dec., 21 win.
`hasArea`	0.940	**0.031**	0.938	0.01 dec., 0 win.

Table 2: Permutation control over 400 shuffles.

That reading is wrong, and the rest of this section is the correction. We report it in full because we believed it for most of the campaign and spent most of our effort on it.

6.5 Coverage is not knowledge

A coverage number asks whether any draw lands within the grader's 5 percent tolerance of the gold. That question has a confound. A pool of one hundred guesses that spans one decade contains roughly twenty one mutually exclusive 5 percent windows, so it will contain something acceptably close to almost any plausible target whether or not the model knows the answer. Coverage therefore measures the width of the guess distribution as well as its correctness, and the two are not separated anywhere in this literature that we are aware of.

The control is a permutation test. Give every subject a different subject's gold value, drawn from the same relation so the null preserves the marginal distribution of plausible answers, and recompute coverage. Whatever survives is coincidence. Four hundred shuffles, validation split:

Both pools carry real knowledge, at z of 12.4 and 53.1 against the null. What differs is how much of the headline number is real. Area draws are close to unanimous, so area coverage is almost entirely knowledge, and rank one selection already realises 0.850 of the available 0.940. Capacity draws are dispersed, and 35.5 of the 84.5 coverage points are the pool being wide enough to hit anything.

Decomposing by where the gold sits in the frequency ranking shows where a selector could actually find something. Excess is the real rate minus the coincidence rate at that rank: at rank 1 it is $0.330 - 0.049 = +0.281$; at ranks 2 to 3, $0.216 - 0.077 = +0.140$; at ranks 4 to 10, $0.206 - 0.160 = +0.046$; at rank 11 or worse, $0.082 - 0.066 = +0.017$.

Rank one is what the shipped selector already emits. The strongly marked knowledge is therefore already realised, and what remains findable is the excess at rank two and beyond: 0.203 of the relation. The raw coverage gap implies 0.443. The recoverable figure is roughly two and a half times smaller than the number we, and the published system we were chasing, had been quoting.

This also explains a fact that had no explanation before. Ranks four and worse are almost entirely coincidence, and no feature can mark a value as correct when its presence in the pool is an accident of spread. That is the mechanical reason why more than twenty published aggregation mechanisms and ten of our own all cap at the same place. They were competing for headroom that is largely not there. We think any oracle style ceiling in closed book knowledge base construction needs this control before it is used to direct effort.

6.6 What did not work, and three faults that cost more

Three selection methods that ought to have helped did not, and the reasoning behind each is the more useful contribution. Cross-frame consensus lost to the best single frame (0.2887 against 0.3196) because the frames share one base model and one prior toward round numbers, so their errors are correlated and they agree on the same wrong attractors. Routing by draw concentration also lost (0.2784 raw, 0.3093 calibrated) because only about a quarter of frame-subject pairs reach support above 0.6, so the cross frame choice is made precisely where no signal exists. And the overshoot scaling reported by a published system did not transfer: all five numeric frames selected a scale of exactly 1.0.

Separately, three implementation faults each cost more than any modelling decision we made, and all three left every self-check green: integer rounding that annihilated sub-unit areas, single-linkage clustering that chained across genuinely different answers, and a leakage guard that excluded only a channel's own demonstrations and so produced an apparent $+0.0788$ gain that was entirely an artifact. Appendix A gives all six in full, with the diagnostics that identified them.

7 Analysis

7.1 Measured priors beat assumed ones

A single all-empty submission is worth more than it looks. Because a row with empty gold and an empty prediction scores 1.0, the per relation macro-F1 of an all-empty file equals that relation's empty-gold rate exactly, so one submission recovers all six test priors.

They are not the validation priors: `personHasCityOfDeath` moves from 39.0% to 48.0%, `companyTradesAtStockExchange` from 38.0% to 44.0%, and `countryLandBordersCountry` from 26.5% to 14.9%. All three exceed a five point shift, and they move in opposite directions. We recentre each threshold by importance weighting validation rows to the measured test rate, worth 0.0064 overall. The assumption is explicit: difficulty within the empty and non-empty groups is unchanged and only their proportion differs, which is weaker than assuming the splits are interchangeable.

7.2 Where the remaining headroom is, and is not

Of the six relations, borders is at its ceiling (0.9942 with a standard error of 0.0033, and a perfect solver would add 0.0008 overall). `companyTradesAtStockExchange` and `hasArea` are within noise of the best posted figures. `awardWonBy` cannot be evaluated at ten rows.

That leaves `hasCapacity`, which holds 20.6 percent of the rows and where the entire field sits below 0.33. We tested our conclusion rather than resting on it. Forced choice duels are the one instrument available to us that is not a transformation of the sampled frequency table. We generated every pairwise comparison among the top six tolerance separated candidates, in both presentation orders, with eight demonstrations balanced so that the correct answer is at position A four times and is the larger value four times. Aggregating by Borda count, with the rule fixed and committed to version control before the pools were generated, the probe scores 0.3196 on validation against the shipped selector's 0.3608. It loses.

The diagnostics matter more than the verdict, because they distinguish a broken instrument from a working instrument reporting an absence. The probe is not degenerate. Position bias is corrected to a mean $P(A)$ of 0.5188, a third of pairs are decided at a margin above 0.25, and on the 580 validation pairs where exactly one side is acceptable the model chooses correctly 62.4 percent of the time, at z of 6.0. The problem is what that accuracy is made of. On the same pairs, always choosing the larger of the two numbers scores 60.3 percent, and the probability the model assigns to the larger option rises monotonically with the ratio between them, from 0.567 below $10^{0.15}$ to 0.727 above $10^{0.75}$. That is the signature of a magnitude prior, not of recall. The comparative knowledge above that prior is 2.1 points at z of 0.7.

instrument	val., vs 0.3608
cross-frame agreement	0.3308 vs 0.3308, 0.0000
forced choice duel	0.3196, −0.0412
yes or no verification	0.2680, −0.0928

Table 3: Three instruments, each asking the model a different question about the same candidate set, all agree.

We then ran the other non-frequency framing, since a duel and a verification have opposite failure modes: a duel must split its probability mass and so inherits any position or magnitude prior, while a verifier can accept everything. The verification probe presents one claim at a time, with balanced yes and no demonstrations whose false claims are drawn from that demonstration subject's own pool, and scores a candidate by the log odds of yes against no at the next token. It is not a yes-bias readout: it accepts only 28.9 percent of candidates at a positive margin, with a median within row margin spread of 0.250. It scores 0.2680 against 0.3608, a difference of 0.0928 whose 90 percent interval excludes zero from below, and it is worse inside both strata of a split-half stability gate. It discriminates, and it discriminates in the wrong direction.

A fourth line of evidence closes it. Sampled frequency is a Monte Carlo estimate of one functional, so the value of estimating it exactly is the infinite draw limit of the accuracy curve. Fitting accuracy as $a - b/n$ across pool depths from 5 to 100 gives a limit of 0.368 to 0.383, a headroom of 0.008 to 0.022 on the relation. Meanwhile two disjoint fifty draw halves of the same pool disagree on 38.1 percent of validation rows at almost no cost in accuracy, because the values they disagree between are equally likely to be wrong.

So the lever is not a better selector, and it is not a better aggregator. On this relation the model is guessing within a plausible range, and the range happens to contain the answer often enough to make coverage look like an opportunity.

Limitations

Most of the numbers in this paper are validation figures with a standard error of about 0.021 on the overall metric and 0.03 to 0.05 per relation. Differences smaller than that, including our apparent margins on `hasCapacity` and `personHasCityOfDeath`, are point estimates and not demonstrated leads. We say so rather than rounding them into claims.

Thresholds are selected using validation, so validation is a selection set and not a clean holdout, and the reported validation figure is optimistic by an amount we estimate at 0.015 to 0.019.

The award relation is not meaningfully evaluable at ten rows and we claim no result on it. Our validation to test transfer estimates rest on a single edition and one published system. The coverage analysis measures whether an acceptable string appears anywhere in a pool, which upper bounds what a selector could achieve; it is a diagnostic, not a method. And the per relation choice of numeric aggregator is supported by a confidence interval excluding zero only for `hasArea`; for `hasCapacity` the interval includes zero and the choice ships as provisional.

Reproducibility statement

Code, prompts, seeds, threshold grids, and the parameter audit are available at https://github.com/thylinao1/lm-kbc-2026. Pools are regenerable from the committed configuration, and the manifest hash for every pool consumed by a submitted predictions file is recorded alongside that file. The full experimental record, including the runs that failed, is in `DECISION-LOG.md` in that repository.

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A Negative results and implementation faults in full

A.1 Three selection methods that did not work

Cross-frame consensus loses to the best single frame. The hypothesis was that a value surviving a change of register is better evidence than one merely modal within a single register. On identical cached draws, consensus scored 0.2887 against 0.3196. The premise was wrong in a specific way: the frames share one base model and one prior toward round numbers, so their errors are correlated. For a venue whose true capacity is 5000, three frames independently proposed 10000, 10500 and 15000. Agreement among correlated predictors certifies nothing.

Routing by confidence also loses. Concentration of draws is strongly predictive of correctness: accuracy rises from 0.000 at support below 0.2 to 0.886 above 0.8. But routing each subject to its most concentrated frame scored 0.2784, and per frame calibrated concentration scored 0.3093, both below the best single frame. The reason is structural: only about a quarter of frame-subject pairs reach support above 0.6, and below that every frame sits near 0.15, so the choice is made precisely where no signal exists. An oracle picking the right frame per subject would score 0.3814. The information is there and three methods failed to extract it; we record that as an open problem rather than a result.

Overshoot scaling does not transfer. A published system reports multiplying capacity predictions by 0.95. On this substrate all five numeric frames selected a scale of exactly 1.0 on both train and validation.

A.2 Three implementation faults

All three were found by reading errors, not by adding compute, and all three are the kind of fault that leaves every self-check green.

Integer rounding annihilated small areas. The numeric writer ended by rounding to an integer, correct for stadium capacities and destructive for areas: golds of 0.43, 0.176, 0.063 and 0.008482 square kilometres all became "0", which can never match. Five validation subjects were lost this way, every one with the correct value already in its pool. Fixing the formatter moved `hasArea` from 0.7600 to 0.8500, a gain of 0.019 on the overall metric, larger than any modelling change we made.

Single-linkage clustering chains. The published aggregation recipe clusters draws in log space and takes the median of the largest cluster. With enough draws, intermediate values bridge distant ones: 10000, 10500, 12000 and 15000 merge into one cluster spanning fifty percent, whose median is a value no draw proposed and no gold at either end accepts. We replaced it with a fixed radius count: score each distinct drawn value by how many draws fall within the grader's own tolerance of it, and take the argmax. A fixed radius cannot chain, and it optimises exactly the quantity the metric rewards.

A leakage guard correct for one channel and wrong for two. Our threshold tuning excludes a channel's own few shot demonstrations from its train curve, since a demonstration subject has its gold answer sitting in its own prompt. That guard is sufficient for any single channel measurement and silently insufficient for any measurement reading two channels at once. Channels draw different demonstration sets from the same 100 train rows, so a subject clean for one frame is frequently a demonstration for another, and a cross frame feature computed on it encodes gold rather than agreement. The union of demonstration subjects is 64 of 100 train rows for `hasCapacity` and 74 of 100 for `hasArea`, reaching all 100 once the 100 shot frame is included, which leaves that relation with no leak free train row at all.

We found this by building a cross frame reranker that keeps the shipped frame as the only proposer of candidates and uses the other frames purely as evidence to reorder them. Pooled over train and validation it read $+0.0788$ on `hasCapacity` and $+0.0476$ on `hasArea`, with a shuffled label control far below and every leave one frame out variant positive. Split by split it was $+0.2206$ on train and exactly $+0.0000$ on validation. On leak free rows only, 133 for `hasCapacity` and 126 for `hasArea`, the effect is 0.0000 with 8 rows up and 8 down, and 2 up and 2 down. Subject grouped cross validation does not catch this and neither does a label permutation control, because both test whether the model is learning something real while the leak is in the feature for that subject. The only check that caught it was asking which split the gain came from.